Great Wall FAQ (v4)

Last updated: August 27, 2026

For Clients (The Problem & Solution)

Q: What is the "$5 wrench attack" and why isn't my hardware wallet safe?

A: The $5 wrench attack is when an attacker threatens you with physical violence to force you to unlock your wallet. Hardware wallets (Ledger, Trezor) protect against remote hackers, not physical coercion: if you are present, you can be forced to unlock. They are, therefore, completely vulnerable to a wrench attack.

Q: How does Great Wall solve this? What are TGPO and TKBA?

A: By making a physical attack futile. Our protocol is **TGPO — the Time-Gated Perceptual Oracle**. It authenticates by *recognition* rather than recall: you recognise something you cannot describe, so there is nothing to hand over under coercion, and a mandated delay means an attacker cannot convert a captive into an immediate transfer. **TKBA (Tacit Knowledge-Based Authentication)** is the property TGPO delivers — four properties that, until now, were mutually exclusive:

1. **Deviceless (Knowledge-Based):** Your mind is your wallet. Access is tied only to tacit knowledge—not a device, seed phrase, or object.
2. **Individual Custody:** We uphold "Not your keys, not your coins." You are always your own bank.
3. **Coercion Resistance:** Access requires time (e.g., 2–168 hours of computation) *and* tacit knowledge (your private gameplay). An attacker has a deficit of time; forced access is impractical.
4. **Anti-Obscurity (Anti-Fragile):** Security increases as attackers learn about it; this knowledge becomes the deterrent.

Q: What is "Loud, Proud, and Free"? Doesn't that make me a target?

A: It's the opposite. In the old paradigm (obscurity), hiding is necessary because your setup is fragile. In the new paradigm (Anti-Obscurity), being loud is a *strategic deterrent*. Our apparel and stickers act like a security company's yard sign—signaling confident protection and deterring attacks before they begin.

Q: Is there one product or two?

A: One mechanism, **two deployments**. The difference is who counts failed attempts:

- **Custodial:** TGPO produces an *authentication decision*. The institution deploys it and imposes the delay through its own system, as it already does for PINs. Because it counts and freezes failed attempts, **$\sim$16 bits** is ample — concretely, **four palettes of sixteen images, reshuffled on every use**, one pick per palette: $16^4 = 2^{16}$, against 10^4 for a four-digit PIN. Since the palette is synthesised server-side and never regenerated from a seed on your hardware, it can use the substrates humans recognise best: **faces and voices**.

- **Self-custodial:** TGPO produces a *derived key*. You deploy it, and the delay is your own time-lock. Nobody counts guesses against an encrypted seed—an attacker copies the ciphertext and attacks it offline, without limit—so this deployment needs **$\sim$160 bits**, derived deterministically on your own device. That is why the self-custodial substrate is a high-entropy one and not a grid of faces.

The gap between 16 and 160 bits is not ambition; it is the absence of a referee.

Q: Which deployment reaches which audience?

A: Neither is a stepping stone to the other; each is the right product for its audience. **Custodial reaches the broad public**—the user adopts nothing, their bank deploys it, exactly as banks deployed transaction limits and hardware tokens before it, so adoption is a procurement decision rather than a conversion. **Self-custodial reaches an earlier-adopting public that already exists**: the community that rolls dice by hand to generate wallet entropy has already demonstrated a tolerance for effortful security ritual that no consumer population shares. The objection that TGPO asks too much of users is answered empirically by the people already doing more.

For Clients (The Product & Service)

Q: What am I actually buying with a subscription?

A: Coercion-proof peace of mind: you outsource the *time* component of self-custodial TGPO via our **Anonymous Computation Marketplace**. Instead of tying up your own machine for hours or days, you pay an anonymous Provider to run the recurring, memory-intensive job.

Q: What are the subscription tiers (Basic, Medium, Professional, Golden)?

A: Tiers map to your security level—the duration of the time-lock computation you pay for:

- **Basic:** 2-hour delay ($1.25/mo)

- **Medium:** 24-hour delay ($18.00/mo)

- **Professional:** 48-hour delay ($42.00/mo)
- **Golden:** 168-hour (7-day) delay ($210.00/mo)

A longer delay implies an attacker would need to sustain a kidnapping for that entire duration—an impractical and futile undertaking.

Q: Is Great Wall a custodian? Do you ever have my keys?
A: No. In the self-custodial deployment we are not a custodian and never hold your keys; TGPO there is a key derivation scheme (like BIP39); we provide the computation marketplace. Only you possess the tacit knowledge to complete access.

Q: What if I forget my tacit knowledge?
A: An integrated memory coach (think Duolingo) helps you practice your private gameplay just enough to retain it without fatigue, creating a healthy habit—and a predictable cadence for your service.

Q: What if the Great Wall company disappears tomorrow?
A: You lose no access. Self-custodial TGPO can run 100% offline on any device. You would only lose convenience—you'd run the hours-long computation yourself.

For Investors (The Business Model & Growth)

Q: What is the Anonymous Computation Marketplace?
A: It's a two-sided marketplace connecting:
Clients: Users who need anonymity and outsource recurring, memory-intensive computation; and
Providers: PC owners (gamers, developers, ex-miners) who monetize idle compute by running those jobs. We take a **28%** platform fee (commission).

Q: Where does the revenue come from, across the whole business?
A: Three segments, counted in three different units and never summed into a single customer number—only annual dollars are comparable across them:

- **S1 — Subscriptions** (B2C marketplace, unit: individuals). The anonymous computation marketplace above. **The only segment included in the Year 1–3 projections.**
- **S2 — Licensing** (B2B, unit: institutions). Custodial TGPO licensed to banks, payment institutions and fintechs, where the *institution* imposes the delay. Annual licence, seats and integration fees only—**no marketplace revenue and no take**

rate, by construction, since an institution that wants the delay to be real never buys capacity to shortcut it.

- **S3 — Capacity sales** (B2B marketplace, unit: companies). Solving capacity sold to companies protecting *digital* assets: treasury keys, signing credentials, root secrets. Same take rate as S1, far higher ARPU, far lower count.

Merchandise is a bonus stream via a separate Superfan Store, outside the primary projections.

Q: What is the Provider-Driven Growth Engine?

A: Providers are our capital-efficient sales force. To head start the network, we offer a **20% lifetime referral commission** on subscription revenue from any client a Provider recruits—*limited to before we reach 20,000 paying customers.*

Q: What are your unit economics (LTV:CAC)?

A: Projected **2̃6:1 LTV:CAC**. CAC is ∼ $21, driven by provider referrals functioning as success-based COGS rather than fixed marketing expense.

Q: What is your current traction?

A: Pre-launch and pre-incorporation, raising a **$600K seed** for **20%** at a **$3M** post-money valuation, against the following milestones:

- MVP Ready: Feb 28, 2026
- First 10 Beta Clients: Mar 10, 2026
- End of Beta (50 Clients): May 2026
- Scale to 1,000 Customers: Sep 2026

Q: What is built today, honestly?

A: A **partial self-custodial prototype** exists. The **custodial deployment is designed but not built**—it is the smaller engineering problem of the two, since it drops the entropy budget, the deterministic derivation and the offline constraint, but it is not yet implemented and nothing here should be read as claiming otherwise.

Q: Why is Client Anonymity so important here?

A: Paying for a time-lock creates a brief *window of vulnerability.* Our marketplace must match Clients and Providers anonymously so attackers cannot learn when a client's window is open. This anonymity also compounds network effects—as users seek to blend into the largest pool.

For detailed economics, see our Executive Summary at loudproudandfree.com.